

# Solving for the toponium Green's function with the method of Jeżabek, Kühn and Teubner

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## Abstract

Artemiy's funky attempt at toponium. Near threshold, everything about a  $t\bar{t}$  pair that feels the QCD potential is contained in one complex function, the S-wave Green's function  $G(E, p)$ . This note shows how I compute it from scratch by following [JKT] equation by equation: the potential of their Sec. 2, the regularisation and numerical solution of their Sec. 3, the checks that the solver is right, and what it gives when used to re-weight MadGraph events. Fuks et al. appear only as the point of comparison: they publish tables of  $G$ , but not how they made them.

## References (PDFs in `reference/`)

- [JKT] M. Jeżabek, J.H. Kühn, T. Teubner, Z. Phys. C 56 (1992) 653 (`BF01474740.pdf`). *The method*.
- [Fuks] B. Fuks, K. Hagiwara, K. Ma, Y.-J. Zheng, arXiv:2411.18962v2 (`2411.18962v2.pdf`). *The comparison*.

Equation numbers always refer to the paper named in front of them: [JKT] (21) is Eq. (21) of JKT. The code is one library, `toponium.py`; a “ $\rightarrow$  name” at the end of a paragraph names the class or method that implements it.

## 1 The problem

A  $t\bar{t}$  pair produced at a point with energy  $E = W - 2m_t$  ( $W$  the pair's invariant mass) and relative momentum  $p$  propagates as

$$G(\vec{p}, E) = \int d^3r e^{i\vec{p}\cdot\vec{r}} G(\vec{r}, 0; E) = \left\langle \vec{p} \left| \frac{1}{\mathcal{E} - H} \right| \vec{r} = 0 \right\rangle,$$
$$H = \frac{p^2}{m_t} + V(r), \quad \mathcal{E} = E + i\Gamma_t \quad ([\text{JKT}] (2)-(4)).$$

The width enters through  $E \rightarrow \mathcal{E}$ : the tops decay before the pair can form a stable bound state, so the toponium “resonances” are smeared into a single bump. Without the potential,

$$G_0(p) = \frac{1}{\mathcal{E} - p^2/m_t} \quad ([\text{JKT}] (6)),$$

and the potential's whole effect is in the ratio  $G/G_0$ . In particular, re-weighting each MadGraph  $t\bar{t}$  event by  $|G(E, p^*)/G_0(E, p^*)|^2$  ([Fuks] (15)) puts toponium into the event sample.

So the task is: *compute  $G(E, p)$  for the QCD potential*. [JKT] give both the potential (their Sec. 2) and a numerical method (their Sec. 3), and I follow both as printed. Where they leave a choice open, I say so and list it in Sec. 7.

**Contrast with Fuks et al.** [Fuks] use the same re-weighting but do not say how they computed  $G$ ; they publish it as a table (swData) and cite [JKT]. My  $G$  is computed independently of those tables. Comparing the two turned out to be a test of the tables rather than of my solver (Sec. 6).

Conventions:  $m_t = 173$  GeV,  $\Gamma_t = 1.49$  GeV,  $C_F = 4/3$ ,  $n_F = 5$ ,  $\alpha_s(m_Z) = 0.12$ .  $G$  has the sign of [JKT] (6). → Constants, Constants.G0

## 2 The method at a glance

| [JKT]     | what it is                                              | in toponium.py                                                            |
|-----------|---------------------------------------------------------|---------------------------------------------------------------------------|
| (14)      | two-loop running coupling $\alpha_s(Q^2)$               | <code>JKTPotential.alpha_2loop,</code><br><code>lambda_msbar_2loop</code> |
| (13)      | perturbative potential $\tilde{V}_{\text{pert}}(Q^2)$   | <code>JKTPotential.V_pert</code>                                          |
| (15)      | full potential: (13) above 5 GeV, Richardson-like below | <code>V_ir, continuity_constant_C,</code><br><code>V_jkt</code>           |
| (19)–(20) | cut at small momentum, and its normalisation            | <code>V_cut, V_cut_position,</code><br><code>energy_shift</code>          |
| (16)–(17) | $G = G_0 K$ and the integral equation for $K$           | <code>LSSolver.G</code>                                                   |
| (18)      | S-wave kernel $\hat{V}(p, q)$                           | <code>LSSolver.kernel</code>                                              |
| (21)–(22) | subtracted equation and $\mathcal{A}(p)$                | <code>LSSolver.script_A, solve_K</code>                                   |
| (23)      | Gauss–Legendre grid                                     | <code>LSSolver.jkt_grid</code>                                            |

|                                                                                                              |  |               |
|--------------------------------------------------------------------------------------------------------------|--|---------------|
| <b>Algorithm.</b> Input: $\alpha_s(m_Z)$ , $m_t$ , $\Gamma_t$ .                                              |  |               |
| 1. Build the potential: $\Lambda_{\overline{\text{MS}}}$ from (14) at $m_Z$ , then (13), (15), the cut (19). |  | Sec. 3        |
| 2. Fix the cut's $\delta^3$ term by (20); it becomes a constant energy shift.                                |  | Sec. 3        |
| 3. Tabulate $F(s) = \int_0^s V dk^2$ , which gives the kernel (18).                                          |  | Step 2        |
| 4. Nodes and weights $q_j, w_j$ (23); matrix $W_{ij} = w_j \hat{V}(q_i, q_j)$ , zero diagonal.               |  | Steps 2, 5    |
| 5. For each $E$ : $g_j = G_0(q_j; E - \text{shift})$ , $\mathcal{A}(q_i)$ (22), solve (21) for $K_i$ .       |  | Steps 3, 4, 6 |
| 6. For each output $p$ : $G(p) = G_0(p) K(p)$ (16), $K$ splined between nodes.                               |  | Step 7        |

## 3 The potential (JKT Sec. 2 and the cut of Sec. 3)

Everything here is in momentum space:  $\tilde{V}(Q^2)$ , with  $Q$  the momentum transfer. → `JKTPotential`

**The running coupling, [JKT] (14).**

$$\alpha_{\overline{\text{MS}}}(Q^2) = \frac{4\pi}{b_0 \ln(Q^2/\Lambda_{\overline{\text{MS}}}^2) + \frac{b_1}{b_0} \ln \ln(Q^2/\Lambda_{\overline{\text{MS}}}^2)}, \quad b_0 = 11 - \frac{2}{3}n_F, \quad b_1 = 102 - \frac{38}{3}n_F.$$

JKT quote  $\alpha_s$  directly; I quote  $\alpha_s(m_Z) = 0.12$ , the value [Fuks] use, and *solve* (14) for  $\Lambda$ : with  $x = \ln(m_Z^2/\Lambda^2)$ , (14) at  $Q = m_Z$  reads

$$b_0 x + \frac{b_1}{b_0} \ln x = \frac{4\pi}{\alpha_s(m_Z)},$$

a single equation in  $x$ , solved by root-finding (Brent's method). This is the only equation solved in building the potential; everything after it is evaluated directly. For  $\alpha_s(m_Z) = 0.12$ ,  $n_F = 5$ :  $\Lambda_{\overline{\text{MS}}}^{(5)} = 223$  MeV, and (14) then gives  $\alpha_s = 0.221, 0.183, 0.157, 0.150$  at  $Q = 5, 10, 20, 25$  GeV.   
→ `beta_coefficients, lambda_msbar_2loop, alpha_2loop`

**The perturbative potential, [JKT] (13).**

$$\tilde{V}_{\text{pert}}(Q^2) = -\frac{16\pi}{3} \frac{\alpha_{\overline{\text{MS}}}(Q^2)}{Q^2} \left[ 1 + \left( \frac{31}{3} - \frac{10}{9} n_F \right) \frac{\alpha_{\overline{\text{MS}}}(Q^2)}{4\pi} \right].$$

This is one-gluon exchange,  $-4\pi C_F \alpha / Q^2$  with  $4\pi C_F = 16\pi/3$ , with two improvements: the coupling runs with the momentum transfer,  $\mu^2 = Q^2$ , by (14), and the square bracket is the one-loop correction to the potential. Nothing is solved: for each  $Q^2$ , compute  $\alpha_{\overline{\text{MS}}}(Q^2)$  from (14), put it into (13). (With  $\alpha$  held fixed this is the potential of [Fuks] (12); without the bracket and with fixed  $\alpha$  it is plain Coulomb, the test case of Sec. 5.1.) → `V_pert`

**Joining to the long-distance potential, [JKT] (15).** (13) is valid only at large  $Q$ ; at small  $Q$  the coupling blows up at  $Q = \Lambda$ . Below  $|\vec{p}| = 5$  GeV JKT switch to a Richardson-like form that also produces the linear confining potential:

$$\tilde{V}(\vec{p}) = \begin{cases} \tilde{V}_{\text{pert}}(p^2, n_F = 5), & |\vec{p}| > 5 \text{ GeV}, \\ -\frac{4}{3} \frac{48\pi^2}{27} \frac{1}{p^2} \left( \frac{1}{\ln(1 + p^2/\Lambda_R^2)} + C \right), & |\vec{p}| < 5 \text{ GeV}, \end{cases} \quad \Lambda_R = 400 \text{ MeV}.$$

$C$  is fixed by continuity at 5 GeV: setting the two lines equal at  $p^2 = 25 \text{ GeV}^2$  and solving for  $C$  (one linear equation) gives  $C = -0.0262$ . At small  $p$ ,  $1/\ln(1 + p^2/\Lambda_R^2) \approx \Lambda_R^2/p^2$ , so  $\tilde{V} \approx -\kappa \Lambda_R^2/p^4$  with  $\kappa = \frac{4}{3} \cdot \frac{48\pi^2}{27}$ : the Fourier transform of a potential rising linearly with  $r$ . → `V_ir`, `continuity_constant_C`, `V_jkt`

**The cut, [JKT] (19).** The  $1/p^4$  behaviour is a problem: its Fourier integral diverges at  $p \rightarrow 0$ , and the kernel (18) built from it is singular as  $(p - q)^{-2}$  at  $q = p$ . JKT argue that  $t$  and  $\bar{t}$  decay before they reach the confining distances, so  $G$  cannot depend on how the potential behaves at very small momentum, and they replace it there:

$$\tilde{V}_{\text{cut}}(\vec{p}; q_{\text{cut}}) = \begin{cases} \tilde{V}(\vec{p}), & |\vec{p}| > q_{\text{cut}}, \\ -\frac{C_1}{p^2 + q_{\text{cut}}^2} + \tilde{V}_0(q_{\text{cut}}) \delta^3(\vec{p}), & |\vec{p}| < q_{\text{cut}}, \end{cases} \quad q_{\text{cut}} = 50 \text{ MeV}.$$

$C_1$  comes from continuity at  $p = q_{\text{cut}}$ :  $-C_1/(2q_{\text{cut}}^2) = \tilde{V}(q_{\text{cut}}^2)$ .

**Normalising the cut, [JKT] (20).** Cutting changes  $V(r)$  by a constant. JKT fix  $\tilde{V}_0$  by demanding that the position-space potential

$$V_{\text{cut}}(r; q_{\text{cut}}) = \int \frac{d^3 q}{(2\pi)^3} e^{-i\vec{q}\cdot\vec{r}} \tilde{V}_{\text{cut}}(\vec{q}; q_{\text{cut}}) = \frac{1}{2\pi^2 r} \int_0^\infty dq q \sin(qr) \tilde{V}_{\text{cut}}(q^2) + \frac{\tilde{V}_0}{(2\pi)^3}$$

satisfy  $V_{\text{cut}}(r = 1 \text{ GeV}^{-1}) = -\frac{1}{4} \text{ GeV}$ . The  $\delta^3$  term is just the constant  $\tilde{V}_0/(2\pi)^3$  in  $V(r)$ , i.e. a constant added to  $H$ , so in the Green's function it is the same as  $E \rightarrow E - \tilde{V}_0/(2\pi)^3$ . I therefore solve with the smooth part of (19) and evaluate  $G_0$  at the shifted energy. For  $\alpha_s(m_Z) = 0.12$  the shift is  $+5.678 \text{ GeV}$ . It is large because it cancels the constant  $\sim \kappa \Lambda_R^2/(2\pi^2 q_{\text{cut}})$  that the cut adds to  $V(r)$ ; it depends on  $q_{\text{cut}}$ , but  $G$  does not (Sec. 5). → `V_cut`, `V_cut_position`, `energy_shift`

## 4 Solving the Lippmann–Schwinger equation (JKT Sec. 3)

**Step 1: the equation, [JKT] (5), (16)–(17).** The Green’s function obeys

$$G(\vec{p}) = G_0(p) + G_0(p) \int \frac{d^3 q}{(2\pi)^3} \tilde{V}(\vec{p} - \vec{q}) G(\vec{q}) \quad ([\text{JKT}] (5)).$$

For the S-wave JKT write  $G = G_0 K$  (16), and the vertex function  $K$  obeys

$$K(p) = 1 + \int_0^\infty dq \hat{V}(p, q) G_0(q) K(q) \quad ([\text{JKT}] (17)).$$

$K$  is smooth in  $p$ : the sharp structure of  $G$  (the on-shell peak at  $p^2 = m_t E$  for  $E > 0$ ) sits in  $G_0$ , which is known exactly.

**Step 2: the S-wave kernel, [JKT] (18).** The angular integral of  $\tilde{V}(|\vec{p} - \vec{q}|)$  gives

$$\begin{aligned} \hat{V}(p, q) &= \frac{1}{4\pi^2} \frac{q}{p} \int_{|p-q|}^{p+q} du u \tilde{V}(u) = \frac{q}{8\pi^2 p} \left[ F((p+q)^2) - F((p-q)^2) \right], \\ F(s) &= \int_0^s dk^2 \tilde{V}(k^2). \end{aligned}$$

$F$  is computed once per potential (10-point Gauss–Legendre on 6000 cells in  $\ln s$ , with the kinks of  $\tilde{V}$  at  $q_{\text{cut}}$  and 5 GeV on cell edges) and splined. This is the only place the potential of Sec. 3 enters the solver. For pure Coulomb,  $\tilde{V} = -B/k^2$ , the integral is elementary:  $\hat{V} = -\frac{Bq}{8\pi^2 p} \ln \frac{(p+q)^2}{(p-q)^2}$ .  
→ `LSSolver.kernel`

**Step 3: the peak at  $q = p$ , and JKT’s subtraction, [JKT] (21)–(22).**  $\hat{V}(p, q)$  is large at  $q = p$  (small momentum transfer). With the cut it is finite but peaked on the scale  $q_{\text{cut}} = 50$  MeV; for Coulomb it has a logarithmic singularity. The grid nodes (Step 5) are  $\sim 0.1$ – $0.25$  GeV apart, so a plain Gauss–Legendre sum misses the peak (Fig. 1, left).

Why not simply put more nodes there? Because “there” is everywhere: the same nodes  $q_j$  serve every row  $p = q_i$  of the linear system (Step 6), and the peak of row  $i$  sits at  $q = q_i$ , on a node of its own. Refining the grid around one  $p$  does nothing for the others; resolving a 50 MeV peak at every node would take a grid  $\sim 100$  times finer,  $N \sim 10^4$ . JKT instead add and subtract  $K(p)$  inside the integral of (17):

$$[1 - \mathcal{A}(p)] K(p) + \int_0^\infty dq \hat{V}(p, q) G_0(q) [K(p) - K(q)] = 1, \quad ([\text{JKT}] (21))$$

$$\mathcal{A}(p) = \int_0^\infty dq \hat{V}(p, q) G_0(q). \quad ([\text{JKT}] (22))$$

This is an identity, not an approximation. It moves the peak into  $\mathcal{A}(p)$ , which contains no unknown: in JKT’s words, “a known function which can be evaluated by a numerical integration”. In the remaining integral the peak is multiplied by  $K(p) - K(q) \approx K'(p)(p - q)$ , which vanishes at  $q = p$ . What is left is  $\sim 1000$  times smaller and odd about  $q = p$ , so the contributions of the nodes on either side largely cancel (Fig. 1, right). The peak is then integrated *once per p*, in  $\mathcal{A}(p)$ , where a fine rule built around that one  $p$  is cheap (Step 4, Fig. 1, middle).

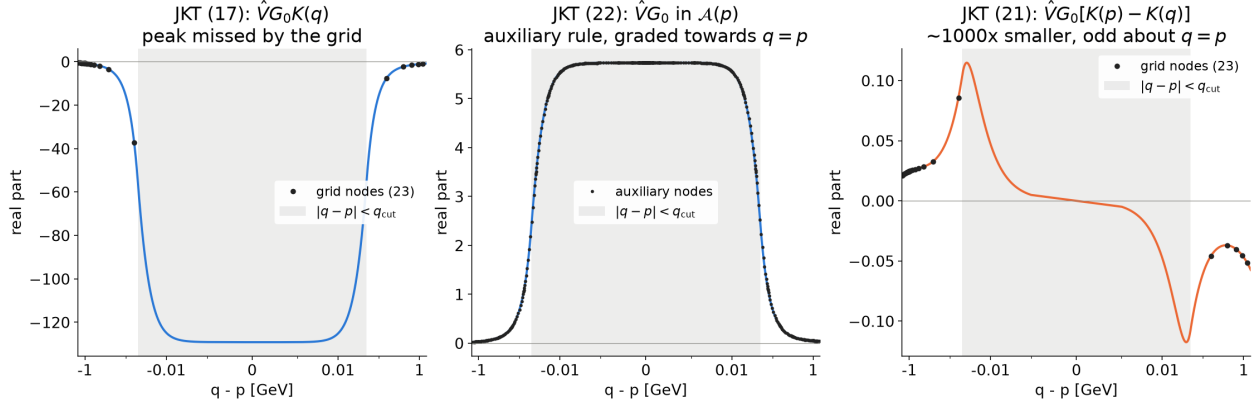


Figure 1: JKT potential ( $\alpha_s(m_Z) = 0.12$ ,  $q_{\text{cut}} = 50$  MeV),  $p = 5$  GeV,  $E = -2$  GeV, against  $q - p$  on a logarithmic scale (linear for  $|q - p| < 1$  MeV) so that the peak region,  $|q - p| \lesssim q_{\text{cut}}$  (shaded), is spread out. Left: the integrand of [JKT] (17) with the grid nodes of [JKT] (23): the peak lies between two nodes. Middle: the integrand of [JKT] (22), on the auxiliary rule of  $\mathcal{A}(p)$ , whose panels are graded geometrically towards  $q = p$  and so are evenly spaced on this axis. Right: the integrand of [JKT] (21),  $\sim 1000\times$  smaller and odd about  $q = p$ , which the grid nodes integrate accurately.

**Step 4: computing  $\mathcal{A}(p)$ , [JKT] (22).** JKT do not say how. I use a fine auxiliary Gauss–Legendre rule for each  $p$ : 16-point panels with edges graded geometrically towards  $q = p$  ( $p(1 \pm 2^{-k})$ ,  $k \leq 24$ , i.e. evenly spaced in  $\ln |q - p|$ ; Fig. 1, middle) to resolve the peak, further edges at  $p(1 + 2^k)$  for its fall-off, every 2 GeV up to 200 GeV for the on-shell peak of  $G_0$ , and a  $t = 1/q$  tail.  $\hat{V}$  on these points is stored once, so for each energy  $\mathcal{A}(p)$  is one dot product with  $G_0$ . Against adaptive quadrature the relative error is  $2 \times 10^{-10}$  (Coulomb) and  $6 \times 10^{-6}$  (JKT, limited by the spline of  $F$ ).  
 $\rightarrow \text{LSSolver.aux\_rule, script\_A}$

**Step 5: the grid, [JKT] (23).** Every  $\int_0^\infty dq$  in (21) becomes a Gauss–Legendre sum  $\sum_j w_j(\cdot)(q_j)$ . JKT use three subintervals,  $(0, a_1)$ ,  $(a_1, a_2)$  and  $(0, 1/a_2)$  in  $t = 1/q$ , with  $a_1 \sim \sqrt{m_t \Gamma_t}$  and  $a_2 \sim m_t$ . I split them once more, at 4 and 60 GeV: edges 0, 4, 16, 60,  $m_t$  plus the  $t = 1/q$  tail, with  $n = 48$  points each ( $N = 240$ ; JKT used  $N = 100\text{--}400$ ). The reason: above threshold at narrow width ( $m_t = 120$ ,  $\Gamma_t = 0.3$  GeV, JKT’s Fig. 4a at  $E = 2$ ), the on-shell peak of  $G_0$  at  $p_0 = \sqrt{m_t E}$  is only  $m_t \Gamma_t / 2p_0 \approx 1.2$  GeV wide. On JKT’s wide middle panel it falls between the nodes, and the peak of  $|pG|^2$  goes  $9272 \rightarrow 7680 \rightarrow 7219$  for  $n = 48, 96, 160$  without settling. With the extra breaks it gives 7065, 7044, 7036 (JKT:  $\sim 7050$ ).  
 $\rightarrow \text{LSSolver.jkt\_grid}$

**Step 6: one linear system per energy.** Imposing (21) at the nodes  $p = q_i$  gives, with  $g_j = G_0(q_j)$  and  $W_{ij} = w_j \hat{V}(q_i, q_j)$ ,

$$\left[1 - \mathcal{A}(q_i) + \sum_{j \neq i} W_{ij} g_j\right] K_i - \sum_{j \neq i} W_{ij} g_j K_j = 1.$$

The  $j = i$  term drops out exactly, because it multiplies  $K_i - K_i = 0$ .  $W$  does not depend on  $E$  and is built once. Each energy costs  $\mathcal{A}$  (Step 4) and one dense  $N \times N$  solve.  $\rightarrow \text{LSSolver.solve\_K}$

**Step 7:  $G$  at any  $p$ , [JKT] (16).**  $G(p) = G_0(p)K(p)$ , with  $K$  between the nodes from a cubic spline in  $\ln q$  through the  $K(q_i)$ . Because  $K$  is smooth and  $G_0$  is exact, this keeps the sharp structure

of  $G$ . (I also tried evaluating (21) itself at off-grid  $p$ . That is 10–40 times less accurate, because at a general  $p$  the nodes are not placed symmetrically around the peak of Step 3.)  $\rightarrow$  `LSSolver.G`

## 5 Is the solver right?

### 5.1 Ground truth: the exact Coulomb solution

For a fixed-coupling Coulomb potential,  $V = -C_F\alpha/r$ , the equation can be solved on paper (JKT mention this below their (12)). The solution is a Whittaker function,

$$G(r) = -\frac{m_t}{4\pi r} \Gamma(1 - \lambda) W_{\lambda, 1/2}(2\kappa r), \quad \kappa = \sqrt{-m_t \mathcal{E}}, \quad \lambda = \frac{C_F \alpha m_t}{2\kappa},$$

and its Laplace representation plus the S-wave Fourier transform give a one-dimensional integral,

$$\tilde{G}(p) = -4m_t\kappa^2 \int_0^\infty dt \frac{t^{-\lambda}(1+t)^\lambda(1+2t)}{(\kappa^2(1+2t)^2 + p^2)^2},$$

which is  $G_0$  at  $\lambda \rightarrow 0$ . It converges at  $t \rightarrow 0$  only for  $\text{Re } \lambda < 1$  ( $\lambda = 1$  is the 1S pole), so it is continued by subtracting  $f(0)$  on  $[0, 1]$ , valid for  $\text{Re } \lambda < 2$ ; since  $|\lambda| \leq 1.08$  for  $\alpha = 0.15$ , one subtraction suffices. No numerical solver is involved.  $\rightarrow$  `CoulombExact`

### 5.2 Accuracy and $q_{\text{cut}}$ independence (Stage 0)

On  $E \in [-6, 4]$  GeV,  $p \in [0.5, 29.5]$  GeV,  $m_t = 173$ ,  $\Gamma_t = 1.49$  GeV. The solver with a Coulomb potential is compared with the exact solution; with the JKT potential, against itself at  $n = 96$ :

| $n$ | $N$ | Coulomb $\alpha = 0.15$ vs exact |                      | JKT $\alpha_s(m_Z) = 0.12$ vs $n = 96$ |                      |
|-----|-----|----------------------------------|----------------------|----------------------------------------|----------------------|
|     |     | median                           | max                  | median                                 | max                  |
| 16  | 80  | $5.3 \times 10^{-5}$             | $1.1 \times 10^{-2}$ | $6.5 \times 10^{-4}$                   | $8.9 \times 10^{-2}$ |
| 24  | 120 | $1.4 \times 10^{-5}$             | $2.2 \times 10^{-3}$ | $2.4 \times 10^{-4}$                   | $2.1 \times 10^{-2}$ |
| 32  | 160 | $5.9 \times 10^{-6}$             | $7.0 \times 10^{-4}$ | $1.2 \times 10^{-4}$                   | $8.9 \times 10^{-3}$ |
| 48  | 240 | $1.8 \times 10^{-6}$             | $2.0 \times 10^{-4}$ | $5.4 \times 10^{-5}$                   | $3.2 \times 10^{-3}$ |
| 64  | 320 | $7.5 \times 10^{-7}$             | $8.0 \times 10^{-5}$ | $2.5 \times 10^{-5}$                   | $1.4 \times 10^{-3}$ |

The maxima sit above threshold on the on-shell ridge; for  $E \leq 0$  they are  $1.1 \times 10^{-5}$  (Coulomb) and  $5.8 \times 10^{-4}$  (JKT) at  $n = 48$ . JKT quote “better than 1%, typically  $10^{-3}$ ”.

$q_{\text{cut}}$  independence (JKT: “ $K$  is independent from  $q_{\text{cut}}$  for  $q_{\text{cut}} < 200$  MeV”). Relative to  $q_{\text{cut}} = 50$  MeV,  $G$  changes by at most  $4.9 \times 10^{-3}$  at 200 MeV,  $1.6 \times 10^{-3}$  at 100 MeV,  $1.0 \times 10^{-3}$  at 20 MeV and  $1.3 \times 10^{-3}$  at 10 MeV, while the energy shift of (20) ranges from 1.6 to 27 GeV. The cut does what JKT claim: it regularises without changing the physics.

### 5.3 Reproducing JKT’s own results (Stage 1)

Since this is JKT’s method, it must reproduce their figures (Fig. 2). I use  $\alpha_s = 0.125$  as JKT do, and  $\Gamma_t$  from [JKT] (8)–(9): 0.299, 0.807 and 1.563 GeV for  $m_t = 120, 150$  and 180 (JKT’s text rounds to 0.3/0.8/1.5). At  $m_t = 120$  the 1S peak of  $R$  is at  $-2.32$  GeV (JKT:  $-2.3$ ) with height 3.80 (JKT: 3.78). The 1S–2S minimum, the 2S bump,  $R(0)$  and  $R(4)$  agree to 0.2–2%. The peaks of  $|pG|^2$  in their Fig. 4a–c agree to 0–3% with the values read off by eye, and  $-G(p \rightarrow 0)$  in their Fig. 2 to 0.4–3%. The exception is  $\text{Im}(-G)(p \rightarrow 0)$  at  $E = 0$  (Fig. 3):  $-2.98$  against their  $-2.7$ , i.e.

10%, on a small value on a  $\pm 8$  axis. The optical theorem [JKT] (12),  $\int \frac{d^3p}{(2\pi)^3} \Gamma_t |G|^2 = -\text{Im } G(0, 0)$ , holds to  $4 \times 10^{-5}$  for  $E \in [-5, 5]$  GeV; it is not built into the solver, so it independently checks the solution.

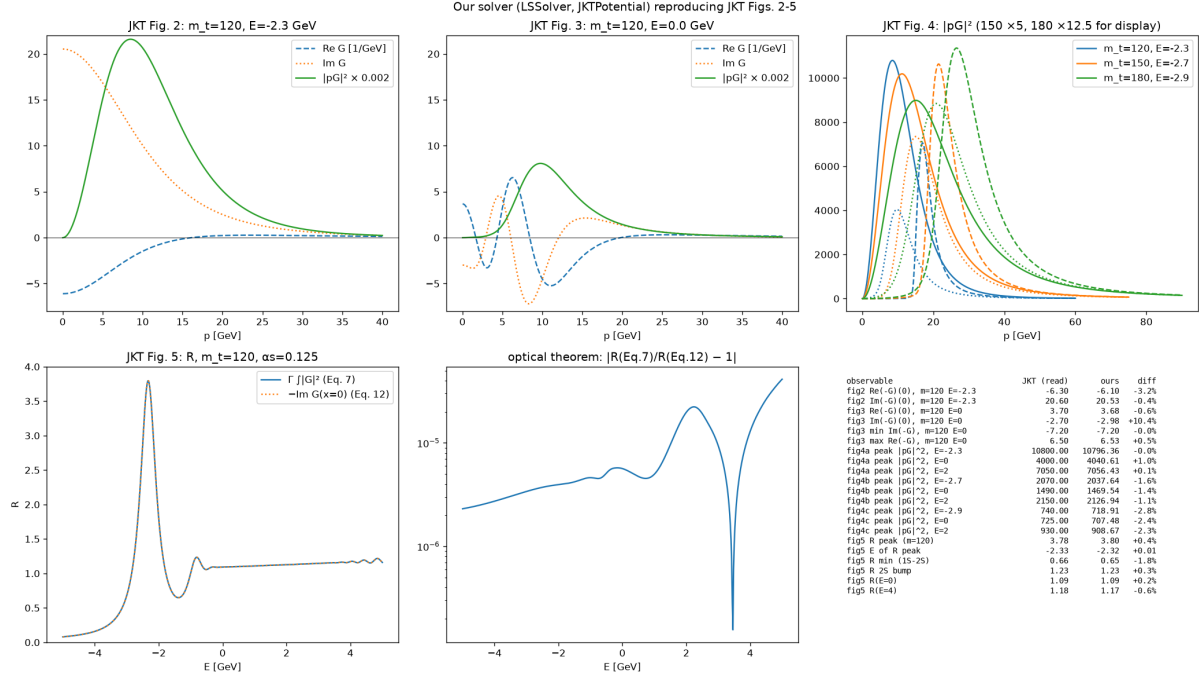


Figure 2: JKT's Figs. 2–5 recomputed with my solver ( $\alpha_s = 0.125$ ,  $\Gamma_t$  from [JKT] (8)). Top:  $G$  and  $|pG|^2$  at  $m_t = 120$  GeV at the 1S peak and at threshold;  $|pG|^2$  for  $m_t = 120, 150, 180$  GeV (solid, dotted, dashed: below, at and above threshold). Bottom: the cross-section ratio  $R(E)$  from  $\int \Gamma_t |G|^2$  and from  $-\text{Im } G(x=0)$  (the optical theorem [JKT] (12)), their relative difference, and the comparison with values read off JKT's figures. (`1.validate_jkt.py`)

One convention to watch: JKT's Figs. 1–3 plot  $-G$  relative to their (6) and (10). At the 1S peak their Fig. 2 has  $\text{Im} \approx +20 \text{ GeV}^{-1}$  at  $p \rightarrow 0$ , whereas  $\text{Im } G = -|\psi(0)|^2/\Gamma_t < 0$ .

## 6 Using it for toponium, and the contrast with Fuks et al.

**What Fuks et al. do.** They re-weight colour-singlet  $gg \rightarrow t\bar{t} \rightarrow b\ell^+\nu \bar{b}\ell'^-\bar{\nu}$  MadGraph events by  $|G/G_0|^2$  ([Fuks] (15)), with  $G$  interpolated from their swData tables by their Fortran routine `CALC GREEN`. They describe the potential behind the tables in four different ways (running tree-level coupling, Richardson following [JKT], fixed  $\alpha_s(m_Z) = 0.12$ , two-loop) and do not describe the method. I read their tables and run their routine exactly as published, as a reference to compare with. → FuksTables

**What is in their tables (Stage 2).** Fitting each of their descriptions to swData, using my solver and the exact Coulomb solution (rms of  $|\text{swData}/\text{model} - 1|$  on  $E \in [-6, 4]$ ,  $p < 30$  GeV):

| Potential                                                                                       | rms    |
|-------------------------------------------------------------------------------------------------|--------|
| fixed- $\alpha$ tree Coulomb, best fit $\alpha = 0.1500$ , $m_t = 172.96$ , $\Gamma_t = 1.4907$ | 0.13%  |
| fixed $\alpha = 0.12$ (their Fig. 1 caption), with or without the bracket of (13)               | 40–50% |
| JKT potential (13)–(15), $\alpha_s(m_Z) = 0.12$ , tree or NLO                                   | 55–69% |
| the same, with a free energy offset                                                             | 15–17% |

So swData is the exact fixed-coupling Coulomb Green’s function at  $\alpha = 0.15$ , not the JKT potential. A plausible origin:  $\alpha_s(m_Z) = 0.12$  run with (14) reaches 0.150 at  $Q \approx 25$  GeV, close to the Bohr momentum (Sec. 3). The authors state neither. My solver with Coulomb  $\alpha = 0.15$  reproduces their tables as well as the exact solution does (median deviation  $2.6 \times 10^{-3}$  for both, Stage 3, Fig. 3), which is a further check of the solver.

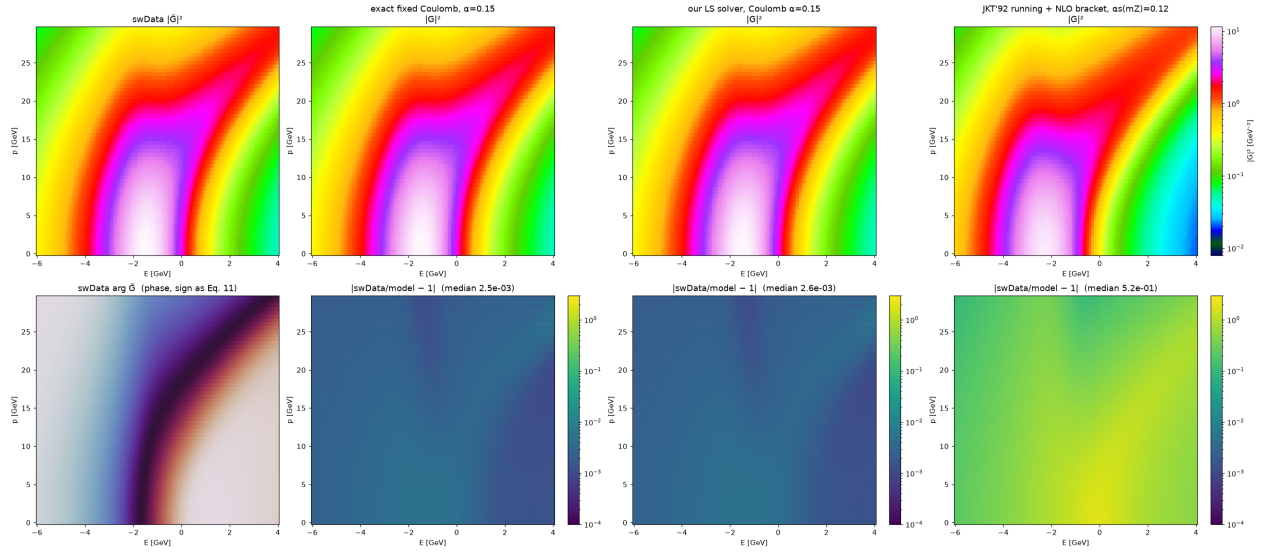


Figure 3: Fuks’s Fig. 1, rebuilt from their swData (left column:  $|\tilde{G}|^2$  and phase), next to exact Coulomb at  $\alpha = 0.15$ , my solver with Coulomb at  $\alpha = 0.15$ , and my solver with the JKT potential at  $\alpha_s(m_Z) = 0.12$ . Bottom row:  $|\text{swData}/\text{model} - 1|$ . (3\_fuks\_fig1.py)

**Events (Stage 4).** 13 TeV, CT18NLO,  $m_t = 173$ ,  $\Gamma_t = 1.49$ : 500k colour-singlet ( $330 \leq W \leq 350$  GeV, colour matrix [Fuks] (20)) and 500k full colour ( $340 \leq W \leq 350$ ); the window is imposed by restricting  $\tau = \hat{s}/s$  in MadGraph’s `genps.f`. The singlet fraction is 0.281 (expected 2/7). The same events are re-weighted with three sources of  $G$ :

|                                               | Fuks: swData<br>via CALCGREEN | Coulomb $\alpha = 0.15$<br>exact | JKT potential (this note)<br>$\alpha_s(m_Z) = 0.12$ |
|-----------------------------------------------|-------------------------------|----------------------------------|-----------------------------------------------------|
| $\sigma_{\text{rw}}/\sigma$ ( $p^* < 50$ GeV) | 8.88                          | 8.90                             | 7.89                                                |
| $\langle p^* \rangle (E = -2)$ [GeV]          | 20.7                          | 20.7                             | 19.6                                                |
| peak of $d\sigma/dW$ [GeV]                    | 344.75                        | 344.75                           | 343.75                                              |

With Fuks’s own  $G$  the re-weighted distributions reproduce their Figs. 2–5 (Figs. 4–7). The JKT potential, i.e. the physics they cite, gives an enhancement smaller by 11%, a peak 1 GeV lower and softer  $p^*$ . → Events

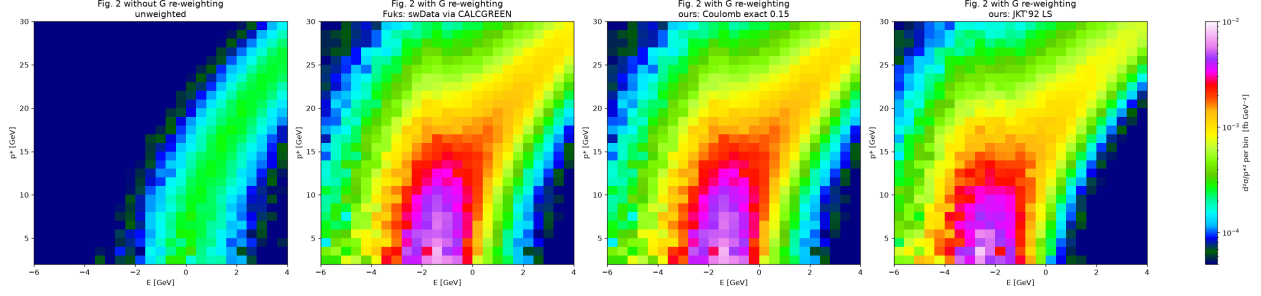


Figure 4: Fuks's Fig. 2 from our events:  $d^2\sigma/p^2$  per bin in the  $(E, p^*)$  plane, without re-weighting and re-weighted with Fuks's  $G$ , exact Coulomb at  $\alpha = 0.15$ , and the JKT potential. (4.fuks\_figs2to5.py)

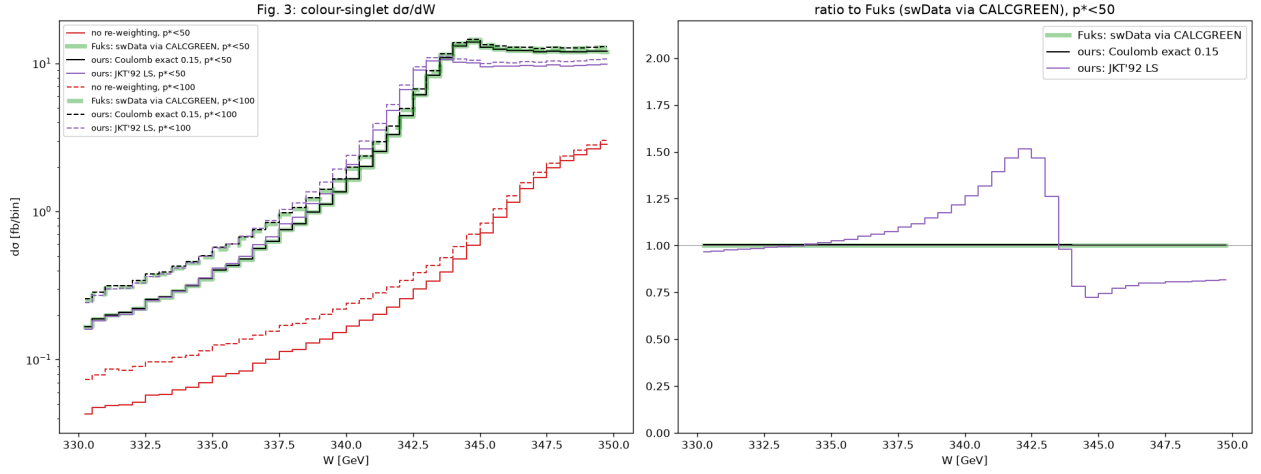


Figure 5: Fuks's Fig. 3: colour-singlet  $d\sigma/dW$  in 0.5 GeV bins, without re-weighting and re-weighted with the three sources of  $G$ , for  $p^* < 50$  and  $p^* < 100$  GeV; right, the ratio to Fuks's own  $G$ . Exact Coulomb at  $\alpha = 0.15$  lies on top of Fuks; the JKT potential moves the peak down by about 1 GeV.

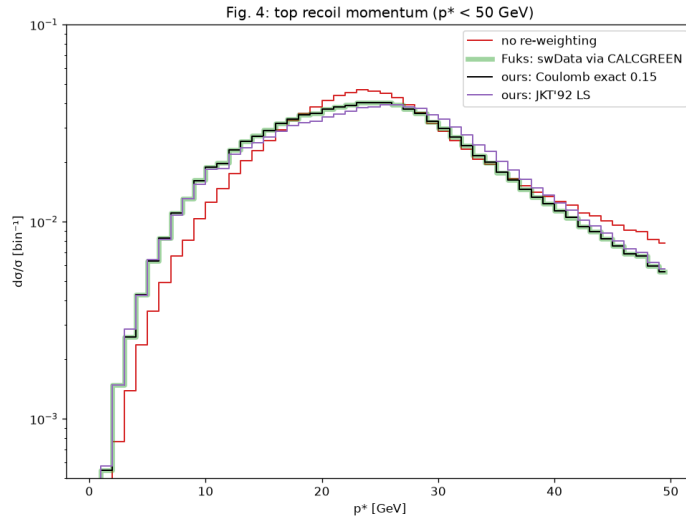


Figure 6: Fuks's Fig. 4: the distribution of the top recoil momentum  $p^*$  ( $p^* < 50$  GeV, fraction per 1 GeV bin), without re-weighting and re-weighted with the three sources of  $G$ .

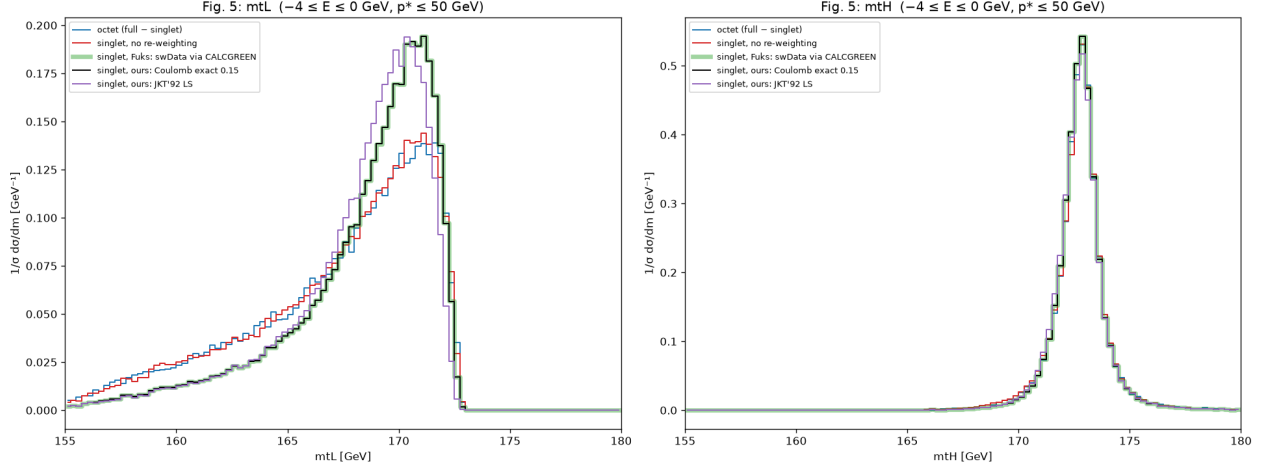


Figure 7: Fuks’s Fig. 5: the lighter and heavier reconstructed top mass,  $m_{t_L}$  and  $m_{t_H}$ , for  $-4 \leq E \leq 0$  GeV and  $p^* \leq 50$  GeV: colour octet, singlet without re-weighting, and singlet re-weighted with the three sources of  $G$ .

## 7 Where I fill in or depart from JKT

1.  $\alpha_s$  is specified at  $m_Z$  and  $\Lambda_{\overline{\text{MS}}}$  obtained by solving (14) there; JKT quote  $\alpha_s$  directly.
2. Grid: JKT’s subintervals split once more, at 4 and 60 GeV (Step 5).
3.  $\mathcal{A}(p)$  on a graded auxiliary Gauss–Legendre rule (Step 4); JKT only say “numerical integration”.
4.  $K$  between the nodes by cubic-spline interpolation (Step 7); JKT do not say how they evaluate off the grid.

Everything else (the potential, the cut and its normalisation, the subtraction, the linear system) is [JKT] as printed.

## A The code, file by file

Everything lives flat in `toponium_study/`. `toponium.py` is the only library; `0...5_*.py` are the six scripts you run (`uv run python 0_solver_checks.py`, and so on), each writing `results/<stage>/summary.txt` and figures. Every class opens with an “EQUATION MAP” listing which paper equation each method implements.

**toponium.py: six classes, top to bottom**

**Constants**

$m_t$ ,  $\Gamma_t$ ,  $C_F$ ,  $n_F$ , and  $G_0$ , [JKT] (6).

**JKTPotential**

Sec. 3: `alpha_2loop`, `lambda_msbar_2loop` (14); `V_pert` (13); `V_ir`, `continuity_constant_C`, `V_jkt` (15); `V_cut` (19); `V_cut_position`, `energy_shift` (20). `JKTPotential.coulomb(alpha)` gives fixed-coupling Coulomb with the same interface, to test the solver.

**LSSolver**

Sec. 4: `kernel` (18), `jkt_grid` (23), `script_A` (22), `solve_K` (21), `G` (16). Usage: `LSSolver(potential).G(E, p)`.

**CoulombExact**

Sec. 5.1: the exact Coulomb  $G$  (`G`, fast; `G_adaptive`, slow reference).

## FuksTables

Fuks's swData and CALCGREEN, used as-is (reference only; nothing of mine is in it).

## Events

the MadGraph events, read from `results/4_fuks_figs2to5/_cache_*.npz`, and the weight  $|G/G_0|^2$  for  $G$  from any of the above.

## The six stage scripts

| script                              | what it does                                                                                                                                                                                                   |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>0_solver_checks.py</code>     | solver vs exact Coulomb, convergence, $q_{\text{cut}}$ independence (Sec. 5);<br><code>kernel_subtraction.png</code>                                                                                           |
| <code>1_validate_jkt.py</code>      | solver vs JKT's own Figs. 2–5; <code>jkt_figs.png</code>                                                                                                                                                       |
| <code>2_identify_swdata.py</code>   | which potential is in Fuks's tables (Sec. 6)                                                                                                                                                                   |
| <code>3_fuks_fig1.py</code>         | Fuks's Fig. 1 rebuilt from swData, my $G$ alongside                                                                                                                                                            |
| <code>4_fuks_figs2to5.py</code>     | events re-weighted with the three sources of $G$ . Reads <code>_cache_full.npz</code> ,<br><code>_cache_singlet330.npz</code> : <b>the only copy of the events, never delete them</b>                          |
| <code>5_comparison_tables.py</code> | CSV tables for comparing with an independent solver (colour singlet): $\Lambda$ , $C$ , $C_1$ ,<br>$\tilde{V}_0$ per setting; $\alpha_s(Q^2)$ and each piece of the potential vs $Q^2$ ; $G$ on $(E, p)$ grids |

`reference/` holds both PDFs and swData; `README.md` has the recipe to regenerate the MadGraph samples; `pyproject.toml/uv.lock` define the Python 3.11 environment.